

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Zoology

Paper: ZOB-301 — Basic Genetics and Evolution and Developmental Biology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Basic Genetics and Evolution) (Marks: 35)

Question 1. Answer the following parts:

- (a) Select and write the best option from the following: $[4 \times \frac{1}{2} = 2]$
- (i) A cross was made between three unlinked genes, $AaBbCc \times AaBbCc$. What would be the frequency of $AABBCC$ individuals?
 - (ii) 1/64
 - (iii) 1/32
 - (iv) 1/16
 - (v) 1/8
 - (vi) The phenomenon of ‘independent assortment’ is associated with:
 - (vii) linked genes
 - (viii) inheritance of genes located on different chromosomes
 - (ix) monohybrid cross
 - (x) location of chromosomes in an interphase cell
 - (xi) If the blood group of the mother is A and the blood group of the father is B, then the blood group of children could be:
 - (xii) A, AB only
 - (xiii) A, B, AB, O
 - (xiv) B, AB only
 - (xv) O, A, B only
 - (xvi) Which of the following statements describes Epistasis?
 - (xvii) An allele that changes the genotype of another allele

- (xviii) A gene that changes the genotype of another gene
- (xix) A gene that masks the expression of another gene
- (xx) An allele that masks the expression of another allele
- (b) Write whether the following statements are True or False giving reasons in 2–3 sentences each: [2 × 2 = 4]
- (i) Darwin finches are examples of adaptive radiation.
- (ii) Monohybrid crosses revealed the phenomenon of incomplete dominance.
- (c) Differentiate between penetrance and expressivity. [2]
- (d) Match the terms in Column-X with Column-Y: [6 × $\frac{1}{2}$ = 3]

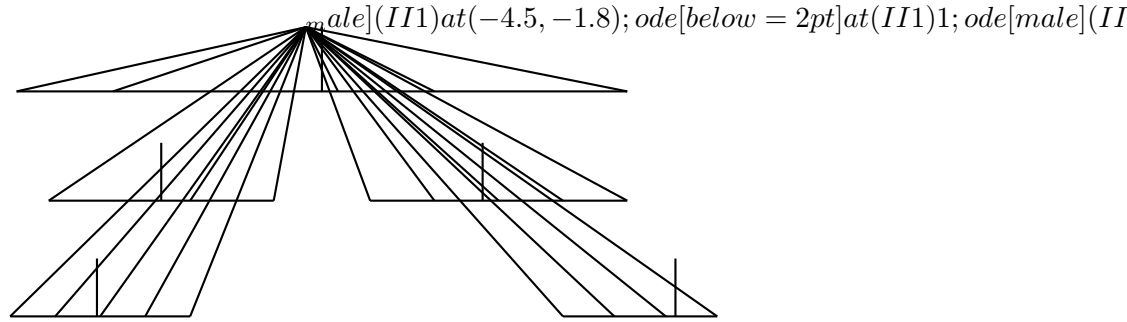
Column-X	Column-Y
(i) Pericentric inversion	(1) Phenocopy
(ii) Wing of a bird and wing of an insect	(2) Coiling of snail shell
(iii) Remnant of ancient animals	(3) Epistasis
(iv) Natural selection	(4) Analogous organ
(v) Sodium metaborate	(5) Chromosome
(vi) Maternal inheritance	(6) Translocation
	(7) Darwin
	(8) Fossil

Question 2. Answer the following parts:

- (a) “Increase and decrease in individual chromosome number in an individual can cause syndromes.” Justify the statement with examples. [4]
- (b) Differentiate between the evolutionary theories of Lamarck and Darwin. [4]
- (c) One variety of plant having leaves with cuts on them was crossed with plants having normal leaves. All the F₁ progeny produced cut leaves. Two of the F₁ plants were interbred to produce F₂. The F₂ had 248 cut leaves and 16 normal leaves. Describe the phenomenon that determines the leaf pattern. [4]

Question 3. Answer the following parts:

- (a) The pedigree represents the pattern of inheritance. Look at the pedigree and answer the following: [6]
- (i) Which type of inheritance is it, dominant or recessive? Justify.
- (ii) Is it autosomal or sex-linked? Justify.



(b) Write short notes on the following:

[2 × 3 = 6]

- (i) Antibiotic resistance in bacteria
- (ii) Chromosomal basis of sex determination

Question 4. Answer the following parts:

- (a) What is Neo-Darwinism and Modern Synthesis of Evolution and how it has redefined the old theory? [4]
- (b) Chromosome 1 of an individual undergoes paracentric inversion. What will be the consequences of meiosis in case crossing-over occurs in the inverted part of chromosomes in a heterozygous condition? Write the answers with the help of diagrams. [6]
- (c) A box has 8 red, 7 blue and 6 green balls. If one ball is picked up randomly, what is the probability that it is neither red nor green? [2]

SECTION – B
(Developmental Biology) (Marks: 35)

Question 1. Answer the following parts:

(a) Define the following: [5 × 1 = 5]

- (i) Regeneration blastema
- (ii) Morula
- (iii) Spermiogenesis
- (iv) Koller's sickle
- (v) Yolk plug

(b) Differentiate between the following: [3 × 2 = 6]

- (i) Preformation and Epigenesis
- (ii) Radial cleavage and Spiral cleavage
- (iii) Autonomous specification and Syncytial specification

Question 2. Answer the following parts:

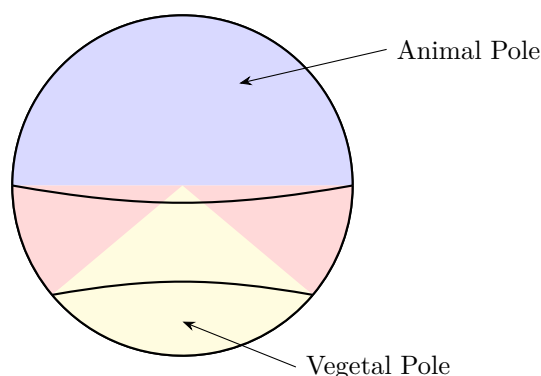
(a) Give an illustrated account of external fertilization in sea urchin. [10]

(b) Write a brief note on the functions of egg membranes. [2]

Question 3. Answer the following parts:

(a) Describe the process of three germ layers formation in chick. [9]

(b) Draw a well-labelled diagram of the fate map of a frog embryo. [3]



Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Molecular responses to thyroid hormones during metamorphosis
- (b) Role of Nieuwkoop center in the formation of dorsal-ventral axis in frog
- (c) Types of eggs

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